

Karthik Rachamolla

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EDUCATION

University of Cincinnati

Bachelor of Science in Computer Science, Minor in Finance

Anticipated Graduation: May 2028

GPA: 3.90

EXPERIENCE

ATech Training

May 2025 – Aug 2025

Embedded Software Engineer Intern

Walton, KY

- Built a full-stack diagnostic app using the Web Bluetooth API to stream live telemetry from 3 ESP32 device variants, reducing technician setup time from ~15 minutes to under 30 seconds per session
- Designed a real-time vehicle diagnostic interface on the ESP32 microcontroller, enabling sub-second data retrieval and display for 6+ sensor channels, accelerating fault identification by 40%

Motz Engineering

Aug 2024 – Dec 2024

Software Developer Intern (Database Systems)

Cincinnati, OH

- Engineered a C# Project Manager integrated with the Ajera API, cutting data retrieval times by 80% through optimized SQL queries and reducing daily manual lookup overhead for 50+ employees
- Built a second full-stack internal management tool for project ideation to implementation, reducing average time spent locating client deliverables by an estimated 35%
- Refactored 3,000+ lines of inconsistently structured C# into modular services, integrating all internal tools, and cut average task completion time by approximately 7 days based on team feedback

PROJECTS

the-search-thing | Search / Information Retrieval

Jan 2026 – Present

- Built a local-first multimodal semantic search desktop app in Rust and Electron, supporting search across text files, images, and videos from a single search interface
- Implemented a Rust JSON-RPC sidecar paired with a hybrid search engine (BM25) build on HelixDB
- Developed an indexing pipeline using Groq enabling semantic retrieval over local media collections of 1,000+ files

Ivy | Agentic AI

Feb 2026 – Mar 2026

- Built a 6-agent book analysis platform in Python using Azure AI Foundry, cutting continuity review time from days of manual sessions to under 5 minutes per manuscript
- Deployed the full stack to production on Azure Apps, with accurate analysis supporting books exceeding 1,000 pages
- Architected a batch-based timeline agent that carefully processes unstructured data to identify plot-holes and understand the book

Quartz | AI Security & Privacy

Nov 2025

- Built a Chrome extension in JavaScript + Cloudflare Workers that performs real-time prompt scanning across LLM web apps, detecting and anonymizing 8+ classes of PII (names, emails, SSNs, financial data, health info) before prompts are submitted
- Engineered a regex-based entity detection pipeline with per-class anonymization rules, processing prompts in under 50ms at the edge via Cloudflare Workers and reducing manual prompt sanitization effort by ~80%

SKILLS

Languages: Rust, C#, TypeScript/JavaScript, Python, C++

DB Systems: SQL (Postgres, MySQL), NoSQL (MongoDB, Azure Cosmos DB, Helix DB, Gremlin)

AI/ML: LLM pipelines, Agentic AI, RAG, NLP, Computer vision

Cloud & Infra: Azure, Cloudflare Workers, GCP, Docker